

ABSTARCT:

In this project, I am doing how to interface LM35 and Begalebone Black along with its program. Once we successfully interface Begalebone Black and lm35, we will go on to build a temperature display using Begalebone Black and a 16×2 LCD module which constantly monitors temperature around the measurement field/range of LM35 and displays the same on LCD module.

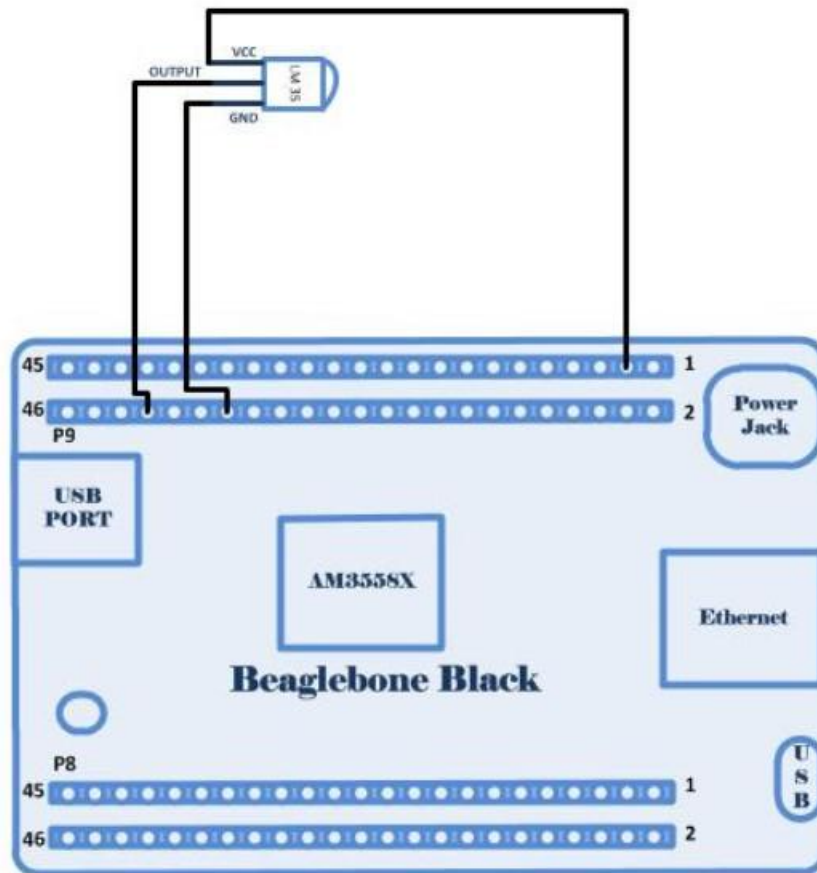
ABOUT LM35:

LM35 is an analog, linear temperature sensor whose output voltage varies linearly with change in temperature. LM35 is three terminal linear temperature sensor from National semiconductors. It can measure temperature from -55 degree Celsius to +150 degree Celsius. The voltage output of the LM35 increases 10mV per degree Celsius rise in temperature. LM35 can be operated from a 5V supply and the stand by current is less than 60uA. LM35 is available in the market in 3 series variations – LM35A, LM35C and LM35D series. The main difference between these 3 versions of LM35 IC is in their range of temperature measurements. The LM35D series is designed to measure from 0 degree Celsius to 100 degree Celsius, where as the LM35A series is designed to measure a wider range of -55 degree Celsius to 155 degree Celsius. The LM35C series is designed to measure from -40 degree Celsius to 110 degree Celsius. In our LM35 Begalebone Black example, we are using the LM35Dz sensor- which falls under LM35D series. So our min-max range of temperature measurement is 0 degree Celsius to 100 degree Celsius.

LM 35 INTERFACES WITH BBB:

LM35 is an analog temperature sensor. This means the output of LM35 is an analog signal. Microcontrollers don't accept analog signals as their input directly. We need to convert this analog output signal to digital before we can feed it to a microcontroller's input. For this purpose, we can use an ADC (Analog to Digital Converter). If we are using a basic microcontroller like 8051, we need to use an external ADC to convert analog output from LM35 to digital. We then feed the output of ADC (converted digital value) to input SOC AM335X. But modern day boards like Begalebone Black and most modern day micro controllers come with inbuilt ADC. Our Begalebone Black has an in built 12 bit ADC. We can make use of this in built ADC of Begalebone Black to convert the analog output of LM35 to digital output.

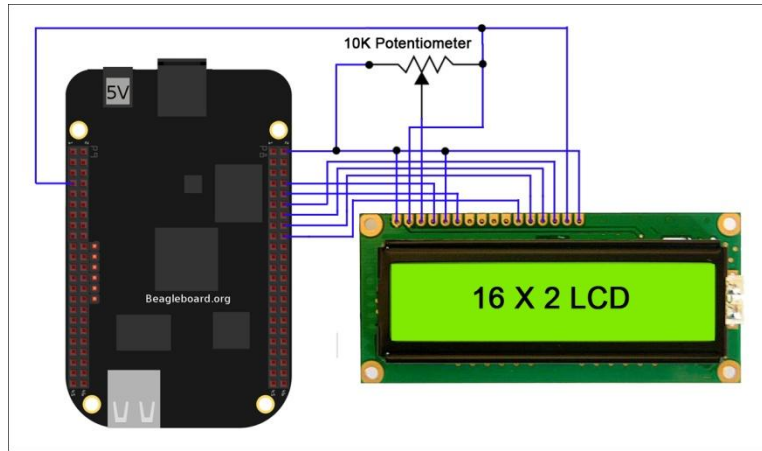
The +5v for LM35 can be taken from the +5v out pin of Begalebone Black . Also the ground pin of LM35 can be connected to GND pin of Begalebone Black . Connect Vout (the analog out of LM35) to any of the analog input pin of Begalebone Black . In this circuit diagram, we have connected Vout of LM35 to **AIN0** of Begalebone Black.



LCD INTERFACE WITH BBB:

Temperature Display on 16×2 LCD Module – using Beaglebone Black and LM35

Now let's go on to add a 16×2 LCD display with LM35 and Beaglebone Black – interface and let's display the temperature values on this LCD display (instead of serial monitor). So we are going to build none other than a standalone temperature display using Beaglebone Black.



Beaglebone Black Pin	LCD Pin
P9_7 (+5V Output)	One outer lead of the potentiometer, LCD pin 2 (VDD) and LCD pin 15 (LED+)
P8_2 (Ground)	Other outer lead of the potentiometer, LCD pin 1 (VSS), LCD pin 5 (R/W) and LCD pin 16 (LED-)
—	Middle lead of the potentiometer to LCD pin 3 (V0/contrast).
P8_8	LCD pin 4 (RS)
P8_10	LCD pin 6 (E/clock enable)
P8_18	LCD pin 11 (DB4)
P8_16	LCD pin 12 (DB5)
P8_14	LCD pin 13 (DB6)
P8_12	LCD pin 14 (DB

ADC:

The BBB has a 12-bit ADC module with a 1.8 V reference. The AM335x, which is the BBB's CPU, uses successive approximation and can take 200,000 samples per second. The digital value can be calculated using the formula.

Analog to Digital Properties

The BBB ADC can sink an maximum of 1.8V.

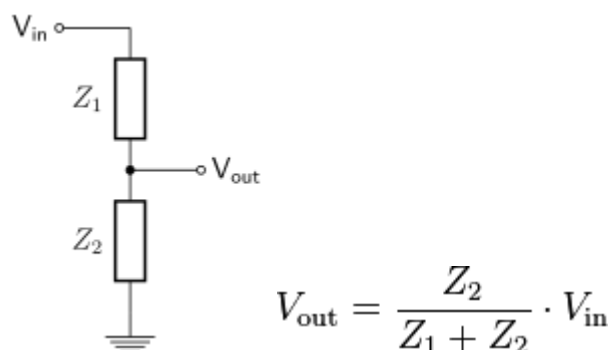


Figure: Voltage divider circuit and equation.

A voltage divider is just a ratio of resistor 2 (Z_2) to the total resistance ($Z_1 + Z_2$). You can think of it in percentages. If we want to cut down from a 5V source to 1.8V, then with a couple resistors we get:

$$V_{in} = 5V$$

$$Z_1 = 1000\ \Omega = 64\%$$

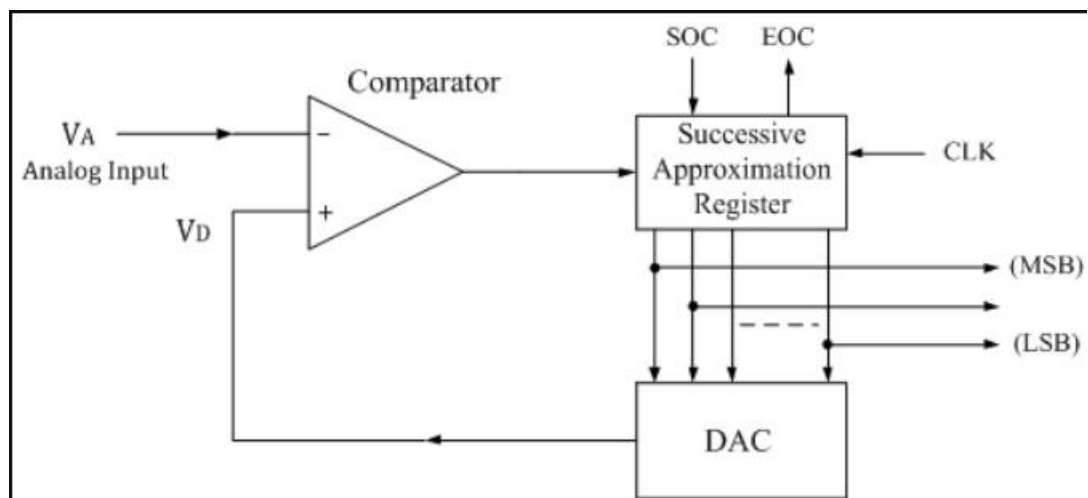
$$Z_2 = 1777\ \Omega = 36\%$$

$$Z_1 + Z_2 = 2777\ \Omega = 100\%$$

Using the equation we find $V_{out} = 1.8V$, which is 36% of V_{in} !

FUNCTIONAL BLOCK DIAGRAM OF ADC:

The functional block diagram of successive approximation type of ADC is shown below.



It consists of a successive approximation register (SAR), DAC and comparator. The output of SAR is given to n-bit DAC. The equivalent analog output voltage of DAC, V_D is applied to the non-inverting input of the comparator. The second input to the comparator is the unknown analog input voltage V_A . The output of the comparator is used to activate the successive approximation logic of SAR.

When the start command is applied, the SAR sets the MSB to logic 1 and other bits are made logic 0, so that the trial code becomes 1000.